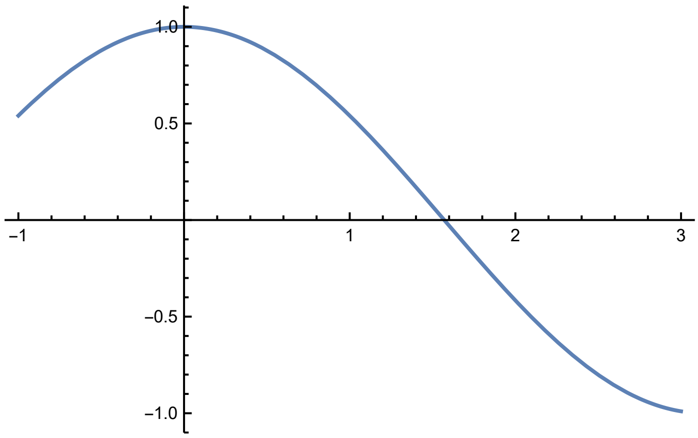
Practical 3(Secant Method)

Question 1

In[ ]:=
```
f[x_] = Cos[x];
Plot[f[x], {x, -1, 3}]
```

Out[ ]=



In[ ]:=
```
x0 = 1;
x1 = 2;
Nmax = 20;
eps = 10^(-6);
Print["f[x]=", f[x]];
For[i = 1, i ≤ Nmax, i++,
  x2 = N[x1 - (f[x] /. x → x1) * (x1 - x0) / ((f[x] /. x → x1) - (f[x] /. x → x0))];
  If[Abs[x1 - x2] < eps, Return[x2], x0 = x1; x1 = x2];
  Print["In", i, "th Number of iterations the root is :", x2];
  Print["estimated error is :", Abs[x1 - x0]]];
Print["root is :", x2];
Print["estimated error is:", Abs[x2 - x1]];
```

```
f[x]=Cos[x]

In1th Number of iterations the root is :1.5649

estimated error is :0.435096

In2th Number of iterations the root is :1.57098

estimated error is :0.0060742

In3th Number of iterations the root is :1.5708

estimated error is :0.000182249
```

Out[ ]=

```
Return[1.5708]
```

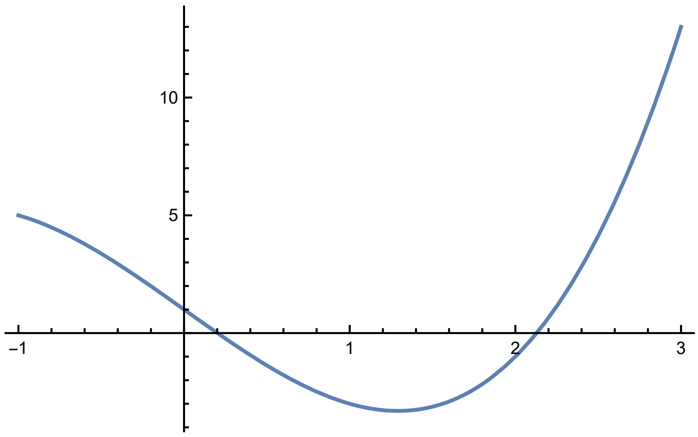
root is :1.5708

estimated error is:1.02185×10^{-9}

Question 2

In[∘]:=
```
f[x_] = x^3 - 5 * x + 1;
Plot [f[x], {x , -1, 3}]
```

Out[∘]=



In[∘]:=
```
x0 = 0;
x1 = 1;
Nmax = 20;
eps = 10^(-6);
Print["f[x]=", f[x]];
For[i = 1, i ≤ Nmax, i++,
  x2 = N[x1 - (f[x] /. x → x1 ) * (x1 - x0) / ((f[x] /. x → x1) - (f[x] /. x → x0))];
  If[Abs[x1 - x2] < eps , Return[x2] , x0 = x1; x1 = x2];
  Print["In", i, "th Number of iterations the root is :", x2];
  Print["estimated error is :", Abs[x1 - x0]]];
Print["root is :" , x2];
Print["estimated error is:", Abs[x2 - x1]];
```

f[x]=$1 - 5x + x^3$

In1th Number of iterations the root is :0.25

estimated error is :0.75

In2th Number of iterations the root is :0.186441

estimated error is :0.0635593

In3th Number of iterations the root is :0.201736

estimated error is :0.0152956

In4th Number of iterations the root is :0.20164

estimated error is :0.0000964033

Out[∘]=

Return[0.20164]

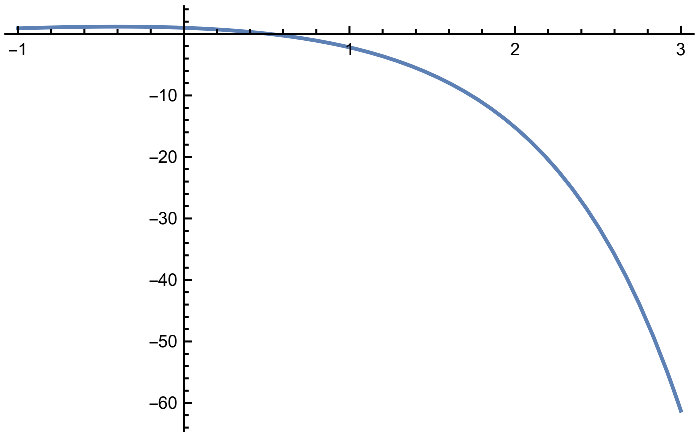
root is :0.20164

estimated error is:1.7717×10^{-7}

Question 3

In[ ]:=
```
f[x_] = Cos[x] - x * Exp[x];
Plot [f[x], {x , -1, 3}]
```

Out[ ]=



In[ ]:=
```
x0 = 0;
x1 = 1;
Nmax = 20;
eps = 10^(-6);
Print["f[x]=", f[x]];
For[i = 1, i ≤ Nmax, i++,
  x2 = N[x1 - (f[x] /. x → x1 ) * (x1 - x0) / ((f[x] /. x → x1) - (f[x] /. x → x0))];
  If[Abs[x1 - x2] < eps , Return[x2] , x0 = x1; x1 = x2];
  Print["In", i, "th Number of iterations the root is :", x2];
  Print["estimated error is :", Abs[x1 - x0]]];
Print["root is :" , x2];
Print["estimated error is:", Abs[x2 - x1]];
```

f[x] = $-e^{x}$ x + Cos[x]

In1th Number of iterations the root is :0.314665

estimated error is :0.685335

In2th Number of iterations the root is :0.446728

estimated error is :0.132063

In3th Number of iterations the root is :0.531706

estimated error is :0.0849777

In4th Number of iterations the root is :0.516904

estimated error is :0.0148014

In5th Number of iterations the root is :0.517747

estimated error is :0.000842998

In6th Number of iterations the root is :0.517757

estimated error is :9.90548×10^{-6}

Out[ ]=

Return[0.517757]

```
root is :0.517757
```

estimated error is:7.07182×10^{-9}